

Utilizing Software for Requirements Management

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Why do projects fail?

The Challenges of Requirement Management.

- Changing client needs (Scope Creep).
- Miscommunication between teams.
- Lost information in long email chains.

The Project Manager's Perspective

- We define requirements to ensure the project meets the client's goals.
- As Project Managers, we focus on functional expectations, not just product features.
- Good requirements management prevents "Scope Creep" (uncontrolled changes).
- We need a single "Source of Truth" for the entire team

User Needs, What do we need from the software?

- **Centralization:** One place for all project demands.
- **Traceability:** Linking a requirement to a specific task or test.
- **Collaboration:** Allowing stakeholders and developers to talk in one place.
- **History Tracking:** Seeing who changed a requirement and why.

The MoSCoW Method

Prioritizing with MoSCoW.

- **Must Have:** Critical for delivery.
- **Should Have:** Important but not vital.
- **Could Have:** Desirable extras.
- **Won't Have:** Out of scope for now.

Functional Specifications & MoSCoW

Functionality Name	Input Information	Output Information	Description of Purpose	Significance (MoSCoW)
Requirements Backlog	User stories, text descriptions, and tags.	A prioritized list of all project tasks.	It centralizes all project needs to prevent information loss.	Must Have
Traceability Matrix	Requirement IDs and Task/Testing IDs.	A report showing the link between needs and results.	It ensures that every single requirement is actually built and tested.	Must Have
Workflow Management	Status updates (To Do, In Progress, Done).	Visual boards (Kanban) and progress charts.	It tracks the real-time progress of each requirement during the project.	Must Have
Version History	Edits to text, date of change, and user ID.	A chronological log of all modifications.	It tracks changes to prevent "Scope Creep" and resolve team misunderstandings.	Should Have
Stakeholder Collaboration	Comments, @mentions, and file attachments.	Email notifications and discussion feeds.	It allows the team to clarify requirements directly within the tool.	Should Have
Reporting & Exporting	Date filters and project status data.	PDF or Excel documents for stakeholders.	It generates official documents to show project status to clients.	Could Have

Comparative Price Analysis

Program Name	Type	Producer	Price (Approx)
Jira Software	Web App	Atlassian	Free (10 users) / \$8.15 user/mo
Trello (Standard)	Web App	Atlassian	Free / \$5.00 user/mo
Azure DevOps	Web App	Microsoft	Free (5 users) / \$6.00 user/mo

Choice

Selected Tool: Jira Software.

Advantages:

- It is the industry standard for IT projects.
- It offers native support for MoSCoW and Agile.
- The free tier allows the whole class to practice.

Price: \$0 for our current lab environment.

Exercise Instructions

Heading: Getting Started

Steps:

1. Go to: atlassian.com/software/jira
2. Click on "Get it free".
3. Log in using your student email or Google account.
4. Create a "Kanban" project named "Lab_Requirements".

Exercise for you!

Task 1: Create 3 User Stories for a "Library App".

Task 2: Assign a MoSCoW label to each story (use labels or priorities), and you should

Task 3: Move one story from "To Do" to "In Progress".

Note: Remember to take screenshots of each step for your final report.